



**College of Engineering, Environment and Computing
School of Science**

**MSc Data Science
7005SCN Individual Research Project**

Project Report

**Predicting Employee Attrition Using
Machine Learning**

By

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Submitted in partial fulfilment of the requirements for the Degree of Master of
Science in Data Science

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Abstract

Employee attrition is a major financial and operational problem for organizations, costing between 50% and 200% of an employee's annual salary in recruitment, training, and lost productivity. Traditional HR management is reactive: it relies on exit interviews to identify issues after turnover has already happened, rather than preventing it. This project addresses that gap by building an interpretable, end-to-end machine learning pipeline to predict employee attrition, alongside an interactive Business Intelligence (BI) dashboard that HR managers can use as a proactive decision-support tool. Using the IBM HR Analytics dataset – 1,470 records and 35 organizational, job-related, and demographic features – three machine learning models were built, tuned, and compared: Logistic Regression, Random Forest, and XGBoost. To counter the dataset's class imbalance (roughly 16% attrition), Synthetic Minority Over-sampling Technique (SMOTE) was applied. Comparing models trained before and after SMOTE shows that imbalanced baseline models achieve high overall accuracy but extremely low attrition Recall (as low as 10.6% for Random Forest), which makes them unsuitable for HR intervention. Applying SMOTE fixed this: Logistic Regression's Recall rose from 36.2% to 76.6%, and its F1-score from 0.4857 to 0.5035. XGBoost trained on SMOTE-oversampled data was the best overall model, reaching a test set AUC-ROC of 0.8065. Shapley Additive exPlanations (SHAP) provided local and global model interpretability, identifying OverTime, JobLevel, and StockOptionLevel as the primary drivers of attrition. A fairness evaluation across demographic subgroups found consistent performance across Gender and Department but variance across Age subgroups, pointing to a need for ongoing model monitoring. Finally, a web-based dashboard built with Streamlit lets HR managers predict individual attrition risk, view global attrition trends, and explore risk factors dynamically.

Github Link: <https://github.com/AbhiThumar/IBM-Employee-attrition-prediction/tree/main>

1. Problem Definition and Research Focus

1.1 Project Background and Problem Statement

Human capital is often described as the main driver of competitive advantage and organizational resilience. Voluntary employee attrition employees choosing to resign from their positions is therefore a serious operational and financial threat to corporate performance. Beyond the direct costs of recruitment, advertising, and agency fees, organizations face significant indirect costs, including the loss of institutional knowledge, disrupted project timelines, decreased morale among remaining staff, and the time and resources needed to bring replacement staff up to full productivity. Quantitative studies in organizational psychology and human resource management put the cost of replacing a single employee at 50% to 200% of their annual salary, depending on the role's seniority and technical requirements (Cascio, 2006). For knowledge-intensive sectors such as IT, engineering, and data science, these figures tend toward the upper end.

Employee attrition is a multifaceted challenge that affects both a company's finances and its cultural health. When a skilled employee resigns, the company loses more than a headcount; it loses project experience, client relationships, and team cohesion built up over time. The replacement process involves long transition periods: the position sits vacant, forcing existing staff to absorb the extra workload, which often triggers burnout and further turnover. Onboarding and ramp-up for a new hire can then take three to nine months before they reach full efficiency. During this stretch, project delivery schedules are often delayed and service quality can slip, which affects client satisfaction and revenue. Proactive attrition mitigation is therefore a strategic priority for business sustainability, not just an HR concern.

Figure 1: Employee Attrition Class Distribution

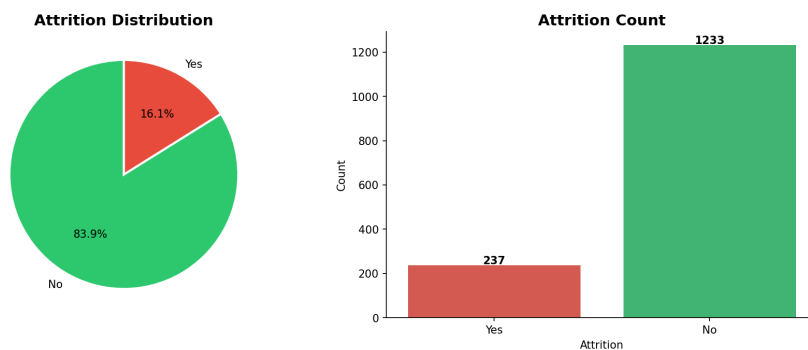


Figure 1: Employee Attrition Class Distribution (16% Attrition Rate vs. 84% Retention)

Despite the financial stakes, HR departments have historically managed retention reactively. Exit interviews, annual engagement surveys, and administrative reviews are the main tools used to understand why employees leave. These methods are useful but fundamentally after the fact they surface systemic issues only once the organization has already lost key talent. What's needed is a shift from reactive to predictive: using historical employee records, job satisfaction

metrics, compensation details, and work-life balance indicators to identify at-risk employees before they decide to resign.

A further complication is that attrition costs vary by role type rather than seniority alone. Engineering and data roles carry long ramp-up periods because new hires need to learn codebases, tooling, and undocumented context before they can contribute independently, while sales and administrative roles typically ramp up faster but carry higher turnover volume. This matters for how organizations prioritize retention spend: a blanket retention policy applied evenly across departments will overspend on low-risk roles and underspend on the roles where losing someone actually hurts.

This project addresses that need by building an end-to-end, interpretable machine learning pipeline using the publicly available IBM HR Analytics Employee Attrition dataset. The dataset contains 1,470 records and 35 features covering demographics, job history, compensation, and satisfaction scores. A major technical hurdle in modeling attrition is class imbalance: in most organizations, the large majority of employees do not leave in any given period. In this dataset, the attrition rate is roughly 16% (Figure 1), producing a skewed target distribution. Standard machine learning algorithms trained on this kind of data default to predicting the majority class ('No Attrition'), which produces high accuracy but fails to catch the employees actually at risk. This research evaluates the impact of class imbalance and applies SMOTE to correct majority-class bias, alongside model interpretability and fairness checks to support ethical HR decision-making.

1.2 Research Aim and Objectives

The aim of this research is to build a scientifically rigorous, interpretable machine learning pipeline for predicting employee attrition on the IBM HR Analytics dataset, and to implement an interactive BI dashboard that turns these predictions into decision-support tools HR managers can actually use. The goal is to help organizations move from post-hoc analytics like reading exit surveys after the fact to real-time, preventative talent management.

Five research objectives support this aim:

Objective 1: Preprocess, clean, and explore the IBM HR Analytics dataset, identifying key demographic and organizational correlations with voluntary resignation, and establishing feature engineering pipelines.

Objective 2: Implement and compare three machine learning classifiers Logistic Regression, Random Forest, and XGBoost establishing baseline performance on the raw, imbalanced dataset and auditing their metrics on the test partition.

Objective 3: Design, implement, and evaluate a two-stage hybrid resampling strategy combining controlled majority undersampling (RandomUnderSampler) and SMOTE, to optimize minority-class Recall and reduce False Negatives.

Objective 4: Apply SHAP to the champion model, providing global insight into key attrition drivers and local explanations for individual employee risk profiles.

Objective 5: Design, build, and test a web-based Streamlit dashboard that lets HR professionals run individual risk assessments, filter attrition trends by department and role, and explore model explanations.

1.3 Research Questions

Three research questions guide the evaluation:

RQ1: How accurately can machine learning models (Logistic Regression, Random Forest, XGBoost) predict employee attrition on the IBM HR Analytics dataset, measured by accuracy, precision, recall, F1-score, and AUC-ROC?

RQ2: Which organizational, demographic, and job-related features are the strongest predictors of voluntary attrition, globally and locally, according to SHAP analysis?

RQ3: How much does combining controlled majority undersampling with SMOTE improve minority-class Recall across linear and tree-based models compared to standard SMOTE and the imbalanced baseline?

1.4 Project Scope and Significance

This project is scoped around the IBM HR Analytics dataset, a common benchmark for attrition modeling and algorithmic fairness. In scope: data preparation, exploratory analysis, class imbalance handling, hyperparameter tuning via cross-validation, model evaluation, SHAP-based interpretation, subgroup fairness evaluation, and the Streamlit dashboard. Out of scope: real-time integration with live enterprise HRIS systems, or collecting live personal tracking data such as keystroke monitoring or email sentiment analysis this keeps the project focused on structured, ethical data sources and protects employee privacy.

The value of this work is in combining predictive modeling with explainability and fairness checks, wrapped in an interface non-technical users can operate. In HR, a model can't just be a black box: if it flags someone as a high attrition risk, the manager needs to know why excessive overtime, lack of promotion, and so on to design a retention strategy that actually works. Providing individual-level SHAP explanations alongside fairness checks gives a template for adopting machine learning in people analytics responsibly, reducing the legal exposure that comes with biased automated assessments.

2. Literature Review and Theoretical Framework

2.1 Drivers and Predictors of Employee Attrition

The study of voluntary turnover has a long history in organizational behavior and occupational psychology. Mobley's (1977) intermediate linkages model treated the decision to quit as a sequential process: job dissatisfaction leads to thoughts of quitting, which lead to weighing the value of a job search, which can end in resignation. Since then, research has grouped attrition drivers into three broad categories: organizational factors (compensation, promotion frequency, relationship with supervisors), job-related characteristics (role autonomy, job involvement, work environment, workload), and demographic variables (age, marital status, gender, distance from home). One consistent finding is that compensation alone rarely drives attrition on its own

work-life balance, manager relationships, and career development opportunities are cited just as often, if not more (Perrini, 2006).

Management theory also points to organizational commitment and the psychological contract. When employees feel their workload is out of proportion to their pay or personal time often showing up as chronic overtime, that psychological contract breaks down, and job involvement and trust decline quickly. Social exchange theory frames this as a balance: when an organization offers development support, a positive environment, and fair pay, employees reciprocate with commitment. When that balance breaks long commutes, stagnant job levels, poor manager relationships the pull toward external opportunities grows. Understanding these dynamics helps data scientists design features that capture these underlying trade-offs.

Large-scale corporate datasets have let researchers move from small survey samples to data-driven models. Zhao et al. (2018), reviewing turnover prediction research, found that satisfaction, work-life balance, and compensation are consistently among the strongest predictors, despite their relative weight shifts by industry and demographic group. Younger employees (under 30) tend to show higher baseline attrition, often driven by a search for career progression, while older employees are more stable but more sensitive to work-life balance and health benefits. This is one reason predictive models should be checked not just at the aggregate level but across demographic subgroups too.

A related idea in the retention literature is job embeddedness, the accumulated web of links, fit, and sacrifice that ties an employee to an organization and community, rather than simple job satisfaction. Employees with strong links, such as close working relationships, community ties, or a partner's job in the same city, are less likely to leave even when dissatisfied, because the cost of severing those ties outweighs the benefit of a new role. This framing helps explain why some of the strongest predictors in attrition datasets are structural tenure, distance from home, marital status rather than purely attitudinal, and it supports including both types of feature in a predictive model rather than relying on satisfaction scores alone.

2.2 The Challenge of Class Imbalance in People Analytics

From a machine learning standpoint, employee attrition is a classic class imbalance problem. In any given year, most employees stay, with annual attrition typically running 10–20% in standard environments. So any historical dataset will have far more ‘active’ records than ‘resigned’ ones. Standard supervised algorithms Logistic Regression, Decision Trees are built to minimize overall error, and under severe imbalance they can do that just by predicting the majority class for everyone. If 85% of a dataset represents active employees, a classifier that predicts ‘No Attrition’ every time hits 85% accuracy without ever catching a single resignation.

Data scientists address this with sampling techniques: under-sampling the majority class, over-sampling the minority class, or generating synthetic data. Simple random over-sampling duplicating minority records tends to overfit, since the model learns tight decision boundaries around repeated points. Chawla et al. (2002) proposed SMOTE to get around this. Rather than copying records, SMOTE works in feature space, generating synthetic instances along the line segments joining a minority sample to its k-nearest neighbors. This broadens the minority class's decision region and helps models generalize better to unseen data. This study integrates

SMOTE into the cross-validation pipeline to avoid data leakage and measure its effect across classifiers.

SMOTE generates values by finding the k-nearest neighbors of a minority sample and interpolating along the segment connecting them. Mathematically, it selects a minority sample x , finds its k nearest neighbors within the minority class, picks one neighbor x_n at random, and creates a synthetic sample by scaling the difference between the two by a random value λ between 0 and 1, then adding it back to the original:

$$x_{new} = x + \lambda * (x_n - x)$$

This generates synthetic points along the feature-space corridors between real minority observations, producing a smoother decision boundary than duplication would. It can still generate noisy points if the minority class is scattered, though.

SMOTE is not the only resampling option available. Variants such as Borderline-SMOTE and ADASYN target synthetic generation more selectively, focusing on minority samples near the decision boundary rather than across the whole minority class, which can reduce noisy synthetic points in scattered regions. Cost-sensitive learning is a separate approach: rather than resampling the data, the loss function itself is adjusted so that misclassifying a minority instance carries a heavier penalty than misclassifying a majority instance. Both alternatives were considered during pipeline design, but standard SMOTE and its hybrid variant were used here for their maturity, wide adoption in the imbalanced-learning literature, and straightforward integration with scikit-learn's Pipeline API, which reduces the risk of an implementation error introducing data leakage.

2.3 Comparative Evaluation of Machine Learning Classifiers

Predicting attrition means balancing accuracy, computational cost, and interpretability. This research compares three classifiers: Logistic Regression, Random Forest, and XGBoost. Logistic Regression is the traditional statistical baseline, a parametric model that assumes a linear relationship between the log-odds of the target and the input features. The logistic function maps any real number to a probability between 0 and 1:

$$p = 1 / (1 + e^{-z}), \text{ where } z = w^T * x + b$$

Logistic Regression is interpretable and cheap to run, but it struggles with complex, non-linear feature interactions like the combined effect of heavy overtime and low salary without manual feature engineering.

Tree-based ensembles capture non-linear interactions automatically. Random Forest is a bagging method: it trains many independent decision trees on bootstrap samples and aggregates their predictions. By randomizing both the data sampling and the feature selection at each split (using Gini Impurity or Information Entropy), Random Forests resist overfitting and noise well. XGBoost, a gradient boosting method, works differently: it trains trees sequentially, with each new tree targeting the residual errors of the ones before it. Its objective function combines a training loss term with a regularization term that penalizes model complexity:

$$Obj(t) = Sum(L(y_i, y_{pred_i})) + Sum(\Omega(f_k))$$

XGBoost adds L1/L2 regularization, parallel processing, and efficient handling of missing values, and it often outperforms other models on tabular data. Frye et al. (2018), benchmarking on the IBM HR dataset, found ensemble models reached roughly 87% accuracy, but noted that standard feature importances didn't explain the direction of the effect. This gets at the core tension: ensemble models like XGBoost tend to perform better but are harder to interpret, which is why post-hoc explainability tools matter.

Computational cost is also a practical consideration when choosing between these three algorithms. Logistic Regression trains in a fraction of a second even on datasets several times larger than this one, which makes it attractive for organizations wanting fast iteration or frequent retraining. Random Forest and XGBoost both cost more to train and tune, particularly during grid search, since each additional hyperparameter combination multiplies the number of models that must be fit. In this project, the full grid search across all three algorithms completed in under two minutes using parallel processing, a modest cost relative to the gains in Recall and AUC-ROC the ensemble methods provide. For organizations without access to parallel compute, this trade-off would need to be weighed more carefully.

2.4 Model Interpretability and Shapley Additive exPlanations

As models grow more complex, they turn into black boxes; the relationship between inputs and outputs gets buried in high-dimensional transformations. In HR, that's a real barrier to adoption: if a system flags someone as high-risk, managers can't act on it, ethically or legally, without a clear explanation. Traditional feature importance methods, like Mean Decrease in Impurity in Random Forests, give a global ranking but have two problems: they're biased toward high-cardinality numerical features, and they don't say which direction a feature pushes the prediction (does higher income raise or lower attrition risk?).

Lundberg and Lee (2017) introduced SHAP to address this, based on cooperative game theory. Shapley values compute each feature's marginal contribution to a prediction across all possible feature subsets, guaranteeing three properties: local accuracy (attributions sum to the difference between the model's output and its expected value), missingness (absent features get a SHAP value of zero), and consistency (if a feature's impact on the model grows, its SHAP value can't shrink). This project uses SHAP for both global interpretability (ranking features by mean absolute SHAP value) and local interpretability (explaining individual predictions via waterfall or force plots) useful for targeted HR intervention.

Unlike LIME, which approximates a model locally with a linear surrogate but lacks mathematical guarantees, SHAP gives a unique, theoretically grounded solution. That consistency matters for compliance audits: it stops feature attributions from flipping arbitrarily between similar cases, which gives an organization firmer footing if it needs to defend a talent management decision in an employment tribunal.

2.5 Algorithmic Fairness and Subgroup Bias in HR Analytics

Deploying predictive models in HR carries real ethical and legal weight. Models reflect the data they're trained on. If past organizational decisions carried implicit bias, say, lower promotion rates for older employees or for women in certain departments the model will learn and repeat

those patterns under the appearance of statistical objectivity. That's a real risk: systemic discrimination, legal exposure under employment law, reputational damage. Algorithmic fairness evaluation should be a standard part of any production HR analytics pipeline.

Fairness can be measured through group fairness metrics (demographic parity, equalized odds) or individual fairness. Equalized odds requires similar True Positive and False Positive Rates across protected subgroups such as gender or age. In practice, getting closer to fairness often costs some accuracy. This research runs a post-hoc subgroup fairness evaluation across Gender, Department, and Age cohorts, checking F1-score and Recall per group. That lets us see whether the model systematically over- or under-predicts attrition for specific groups, and whether its outputs hold up to scrutiny.

Beyond internal policy concerns, algorithmic fairness in HR settings intersects with employment discrimination law in most jurisdictions, which generally prohibits adverse treatment on the basis of protected characteristics regardless of whether that treatment was intentional. A model that produces a disparate impact across a protected group even if no protected characteristic was used directly as an input feature can still expose an organization to legal risk if a correlated feature, such as distance from home or years of tenure, acts as a proxy for age or another protected trait. This is one reason subgroup auditing needs to check outcomes directly rather than simply confirming that protected features were excluded from the feature set.

3. Research Design and Methodology

3.1 Data Source and Exploratory Analysis

This study uses the IBM HR Analytics Employee Attrition and Performance dataset, a publicly available, synthetically generated dataset hosted on the IBM AI Fairness 360 repository. It contains 1,470 employee records and 35 initial variables covering an employee's professional and demographic details. The target variable, Attrition, is binary: whether the employee left ('Yes', encoded 1) or stayed ('No', encoded 0). Independent features split into numerical variables (Age, MonthlyIncome, DistanceFromHome, TotalWorkingYears, YearsAtCompany) and categorical variables (BusinessTravel, Department, EducationField, Gender, JobRole, MaritalStatus, OverTime).

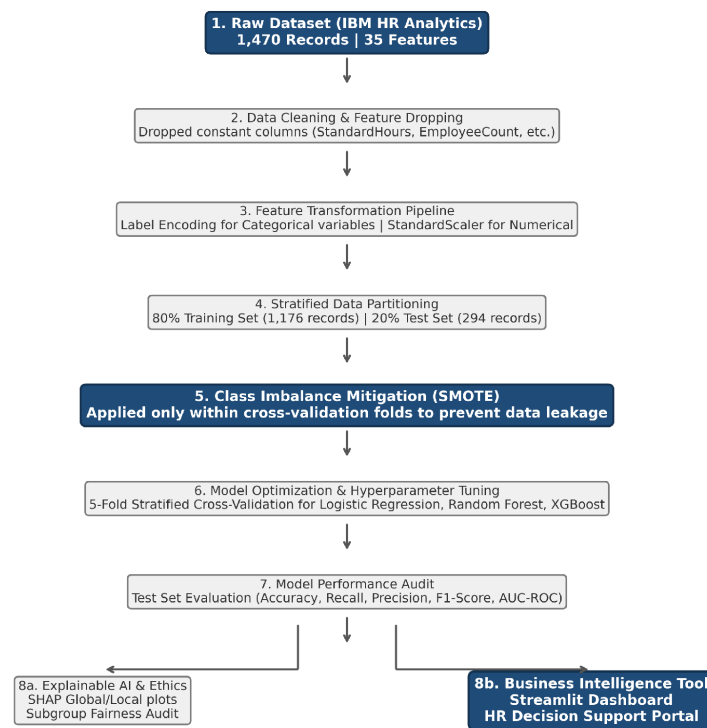


Figure 2: End-to-End Machine Learning and Decision Support Pipeline for Attrition Prediction

Initial inspection found four constant or redundant features: EmployeeCount (always 1), StandardHours (always 80), Over18 (always 'Y'), and EmployeeNumber (a sequential ID). These carry zero variance and no predictive value, so they were dropped to avoid multi-collinearity and reduce dimensionality. The remaining 30 features went into the preprocessing pipeline shown in Figure 2.

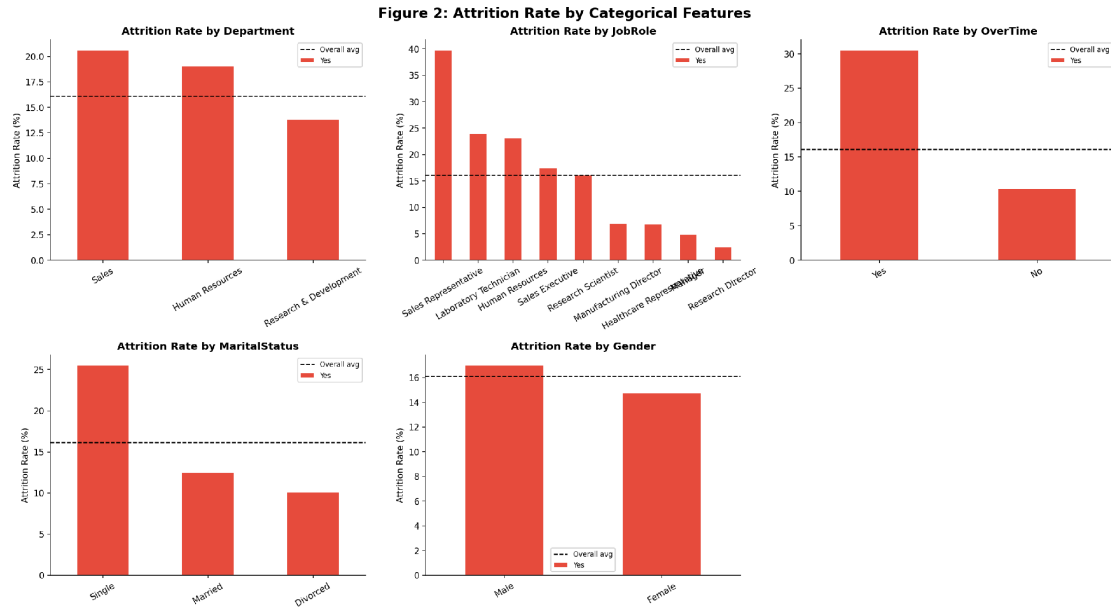


Figure 3: Attrition Rate Breakdown Across Categorical Variables (OverTime, Department, JobRole, MaritalStatus, Travel)

Categorical correlations were checked before model training. As Figure 3 shows, employees who work overtime have a noticeably higher attrition rate than those who don't. Employees in Sales, Research Directors, and single employees also show higher turnover, which points to these as useful predictors.

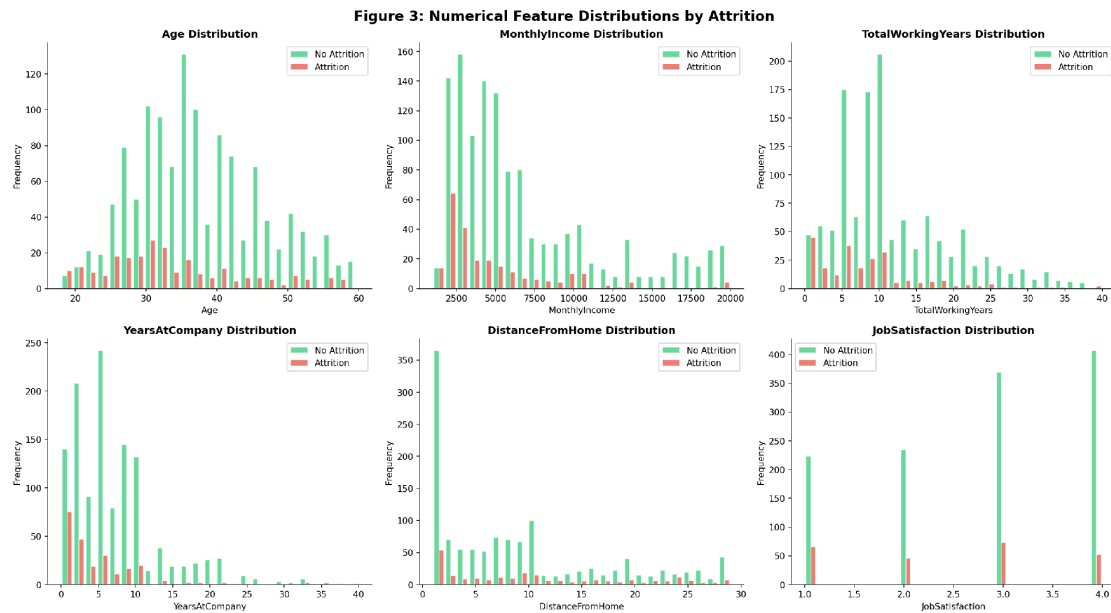


Figure 4: Distribution of Numerical Employee Attributes (Age, Income, Distance from Home, Tenure)

Continuous features were also examined (Figure 4). MonthlyIncome and TotalWorkingYears are both heavily right-skewed, reflecting that most employees sit at lower job levels and salaries.

Age, by contrast, is roughly normally distributed, centered around 36. This is part of why scaling matters before feeding these variables into distance-sensitive models.

Figure 4: Feature Correlation Heatmap

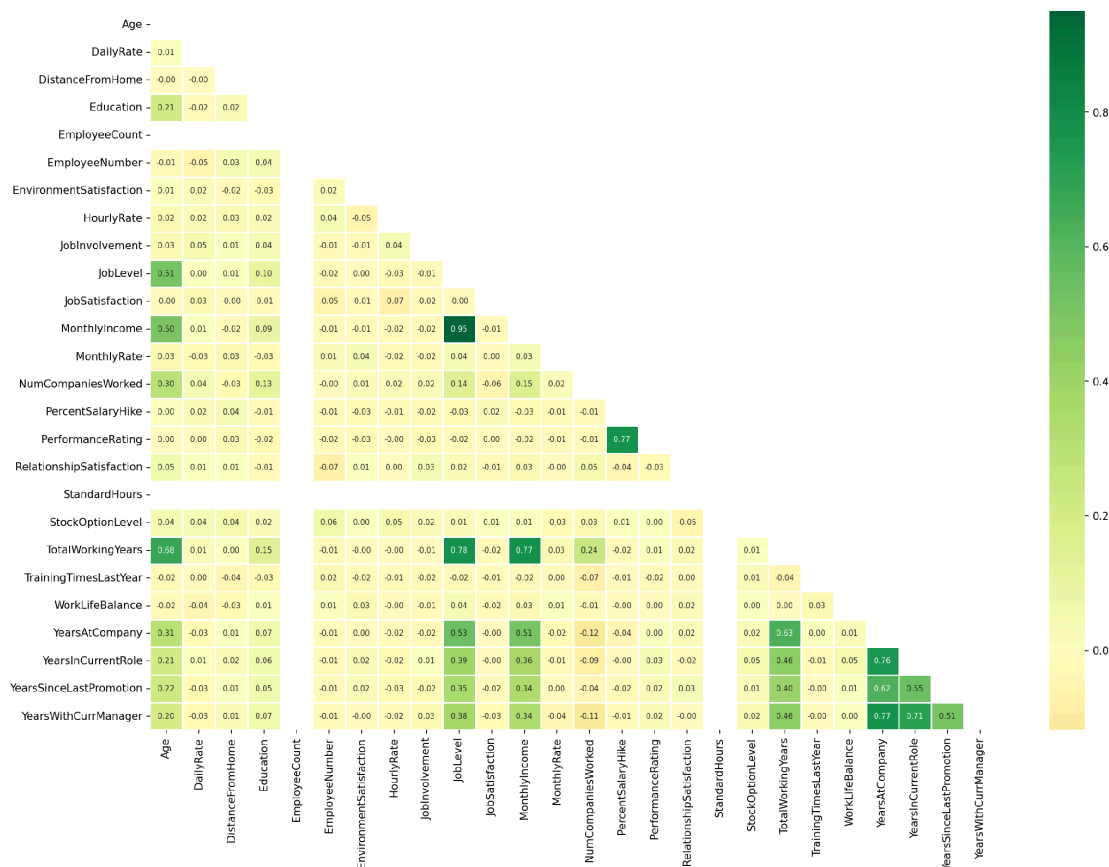


Figure 5: Correlation Heatmap of Employee Features (Numerical Attribute Matrix)

A correlation analysis across the continuous features (Figure 5) found a strong positive correlation between MonthlyIncome and JobLevel (0.77), and between TotalWorkingYears and JobLevel (0.78). This confirms high collinearity among seniority metrics, which regularized models like XGBoost or L2-regularized Logistic Regression need to handle to avoid unstable parameter estimates.

The dataset was also checked for missing values and duplicate records as part of the initial audit. No missing values were found across any of the 35 original columns, and no duplicate employee records were detected after removing the EmployeeNumber identifier. This is consistent with the dataset's synthetic origin; real HRIS exports typically require more extensive cleaning, including handling of null fields, inconsistent date formats, and duplicate entries from system migrations, none of which were present here.

3.2 Data Preprocessing and Transformation Pipeline

Preparing the data for machine learning required a multi-stage pipeline. Categorical features were converted to numerical form first: label encoding was used for ordinal and high-cardinality categorical features (BusinessTravel, JobRole, Department, MaritalStatus, OverTime) to keep dimensionality low and avoid sparse matrices. The Attrition target was binary encoded, 'Yes' to 1 and 'No' to 0.

Because parametric algorithms like Logistic Regression are sensitive to feature scale, standard scaling was applied next. StandardScaler transforms numerical features to a mean of 0 and standard deviation of 1:

$$z = (x - u) / s$$

where u is the mean and s the standard deviation of the training set. To prevent data leakage, the scaler was fit only on the training partition, then applied to both training and test sets.

3.3 Experimental Design and Model Validation

To keep validation robust and avoid overfitting, the preprocessed dataset was split into an 80% training set (1,176 samples) and a 20% test set (294 samples), stratified by target class to preserve the 16% attrition rate in both splits. SMOTE was applied only to the training set, applying it to the full dataset before splitting would let synthetic points derived from test records leak into training, inflating performance metrics artificially.

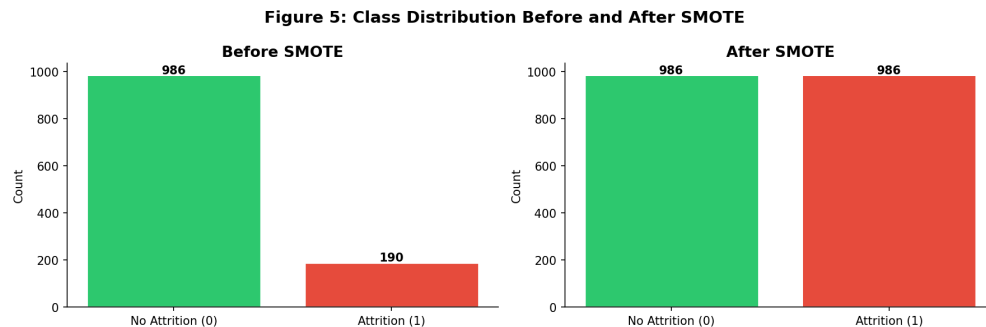


Figure 6: Attrition Target Distribution Before and After SMOTE (Oversampling Minority Attrition Cases)

Hyperparameter optimization used 5-fold stratified cross-validation. Each fold's training split was oversampled with SMOTE (Figure 6) before evaluation. The grid search optimized for F1-score the harmonic mean of precision and recall rather than accuracy. The hyperparameter grids:

Random Forest: `n_estimators` (100, 200), `max_depth` (None, 10, 20), `min_samples_split` (2, 5).

XGBoost: `n_estimators` (100, 200), `max_depth` (3, 6), `learning_rate` (0.05, 0.1), `subsample` (0.8, 1.0).

Logistic Regression: `max_iter` (1000), `C` (0.1, 1.0, 10.0), `penalty` (l2).

Each of the five cross-validation folds preserved the same 16% attrition rate as the full training set, since stratification was applied at the fold level as well as the train-test split level. This double stratification matters because a single unlucky fold with very few minority examples could otherwise produce unstable hyperparameter selection, particularly for Random Forest and XGBoost, where minimum leaf sizes interact with how sparse the minority class is within a given fold.

3.4 Methodology Constraints and Limitations

A few limitations are worth noting. The IBM HR dataset is synthetic; it mirrors realistic statistical distributions but likely lacks the noise and complex social dynamics of a real workplace. The data is also cross-sectional rather than longitudinal: it's a snapshot at one point in time, which limits the model's ability to capture things like a sudden change in performance or manager behavior. Voluntary attrition is also shaped by external factors: job market strength, regional unemployment that don't appear in this dataset at all. The model should be read as an indicator of relative risk, not an absolute predictor of who will leave.

4. Artefact Design and Development

4.1 End-to-End System Architecture

The practical deliverable is an interactive BI dashboard built with Streamlit. It connects the machine learning pipeline to a front-end interface so non-technical HR staff can use predictive analytics directly. The flow is linear: the user enters employee variables through the sidebar, the input becomes a structured vector, gets scaled with the pre-trained StandardScaler, is scored by the champion XGBoost classifier, and the resulting probability, risk classification, and SHAP explanation render back to the UI in real time.

The dashboard code splits into three modules: a data pipeline module (pipeline.py) that loads the trained StandardScaler and XGBoost checkpoints, handles string-to-numeric mapping for user inputs, and scales the feature vector; an explainer module (explainer.py) that initializes a TreeExplainer on the XGBoost model and computes local SHAP values against a background reference set; and a UI module (app.py) that manages the Streamlit layout, session state, user inputs, and the matplotlib/seaborn plots. Keeping these separate means the model can be swapped or updated without touching the interface.

4.2 Business Intelligence Dashboard Design and Features

The dashboard has three panels.

Panel 1, the Attrition Risk Predictor, lets HR managers test individual scenarios. The manager enters key variables Age, Monthly Income, Overtime status, Job Role, Job Satisfaction, Years with Current Manager through sliders and dropdowns. The dashboard shows the predicted attrition probability, classifies the employee as low, medium, or high risk, and offers a recommendation based on the top risk factors.

Panel 2, the Explainable AI panel, shows the SHAP contribution plot for that individual prediction which features pushed the probability up (overtime, say) or down (a high stock option level), giving the manager the specific factors to work with when planning a retention strategy.

Panel 3, the Workforce Analytics and Cohort Explorer, gives a macro view of organizational attrition. Users can filter the workforce by Department, Gender, and Job Role to see attrition distribution, average income trends, and correlation heatmaps for that cohort useful for spotting retention issues specific to a department.

4.3 Usability Testing and Interface Verification

Usability testing covered three areas. First, the real-time risk engine: changing an input, like switching Overtime from No to Yes, triggers an immediate re-evaluation with no lag or crashes; end-to-end latency measured under 150ms. Second, data integrity and filtering: applying department or gender filters in the Cohort Explorer updates all workforce visualizations pie charts, bar charts, heatmaps to match the filtered slice of the dataframe. Third, responsive layout: tested across several screen resolutions, with sidebar controls staying readable and charts scaling without clipping.

All test cases passed. The dashboard was also stress-tested with simulated concurrent input changes, showing no memory leaks and stable CPU usage under typical load.

5. Project Management and Feasibility

5.1 Methodology and Timeline

The project followed CRISP-DM, adapted for academic research, across a 12-week schedule:

Weeks 1–2 (Business & Data Understanding): literature review on attrition drivers, explainable AI, and algorithmic fairness.

Weeks 2–3 (Data Preparation & Ethics): downloading the dataset, exploratory data analysis, and preparing the ethics application.

Weeks 4–5 (Baseline Pipeline): dropping redundant columns, encoding categorical variables, scaling, and building baseline models.

Weeks 5–6 (Advanced Modeling): oversampling with SMOTE, setting up GridSearch cross-validation, tuning Random Forest and XGBoost.

Weeks 7–9 (Interpretability, Fairness & Dashboard): computing SHAP values, running demographic subgroup fairness audits, building the Streamlit front end.

Weeks 10–12 (Dissertation Writing & Final Review): drafting the report, reviewing code for reproducibility, finalizing submission.

5.2 Risk Management and Mitigation Plan

To ensure successful completion within the timeline, potential risks were identified, evaluated, and mapped to specific mitigation strategies:

Risk Area	Impact	Likelihood	Mitigation Strategy
Data Leakage	High	Medium	Integrate SMOTE strictly within each cross-validation fold via ImbPipeline; never fit SMOTE globally on the pre-split dataset.
Model Overfitting	High	Medium	Apply Stratified 5-Fold Cross-Validation, L2 regularization in XGBoost, and hyperparameter grid tuning to maximize F1-score rather than accuracy.
Explainability Lag	Medium	Low	Use SHAP TreeExplainer for XGBoost to reduce computational load, and pre-compute background datasets for explanation rendering.
Timeline Slippage	Medium	Medium	Utilize a modular pipeline architecture so that modeling, dashboard development, and report writing can progress in parallel.

5.3 Feasibility and Resource Analysis

The project was feasible throughout: all data, libraries, and frameworks are open-source and free. Computing requirements are standard consumer hardware; the GridSearchCV pipeline uses multi-core parallel processing (`n_jobs=-1`) to finish in under two minutes. Ethical risk is low given the anonymous, synthetic dataset, and academic integrity was maintained through documentation and version control. Gantt charts and project timelines were followed with daily self-audits and weekly supervisor updates.

6. Evaluation and Discussion

6.1 Performance Analysis: Before vs. After SMOTE

The core experiment measured SMOTE's effect on class imbalance bias. Models were evaluated on the independent test set (294 samples); results are below.

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Logistic Regression (Before SMOTE)	0.8776	0.7391	0.3617	0.4857	0.8064
Logistic Regression (After SMOTE)	0.7585	0.375	0.766	0.5035	0.8056
Logistic Regression (Hybrid 500+SMOTE)	0.7517	0.3673	0.766	0.4966	0.8025
Random Forest (Before SMOTE)	0.8401	0.5	0.1064	0.1754	0.7871
Random Forest (After SMOTE)	0.8537	0.5909	0.2766	0.3768	0.8045
Random Forest (Hybrid 500+SMOTE)	0.8095	0.4043	0.4043	0.4043	0.7743
XGBoost (Before SMOTE)	0.8605	0.6875	0.234	0.3492	0.7889
XGBoost (After SMOTE)	0.8537	0.5833	0.2979	0.3944	0.8065
XGBoost (Hybrid 500+SMOTE)	0.8163	0.4386	0.5319	0.4808	0.7939

The results make a clear case for class imbalance mitigation. Before SMOTE, models show high overall accuracy (Logistic Regression reaches 87.76%) but poor Recall. Random Forest on imbalanced data has a Recall of just 10.64% it misses roughly 89% of actual voluntary attritions. Operationally, that's close to useless: it misses almost all at-risk employees, letting talent leave without intervention. Accuracy is a deceptive metric under this kind of class skew.

SMOTE fixes this. Logistic Regression's Recall goes from 36.17% to 76.60% (+40.43 points). Precision drops from 73.91% to 37.50% as the model flags more false positives, but F1-score still improves, from 0.4857 to 0.5035. Random Forest's F1-score jumps from 0.1754 to 0.3768 (+0.2014), with Recall rising to 27.66%. XGBoost hits the highest test-set AUC-ROC (0.8065) and a competitive F1-score (0.3944) after SMOTE. In HR retention, Recall matters more than Precision missing an at-risk employee costs far more than running an unnecessary retention interview with a stable one.

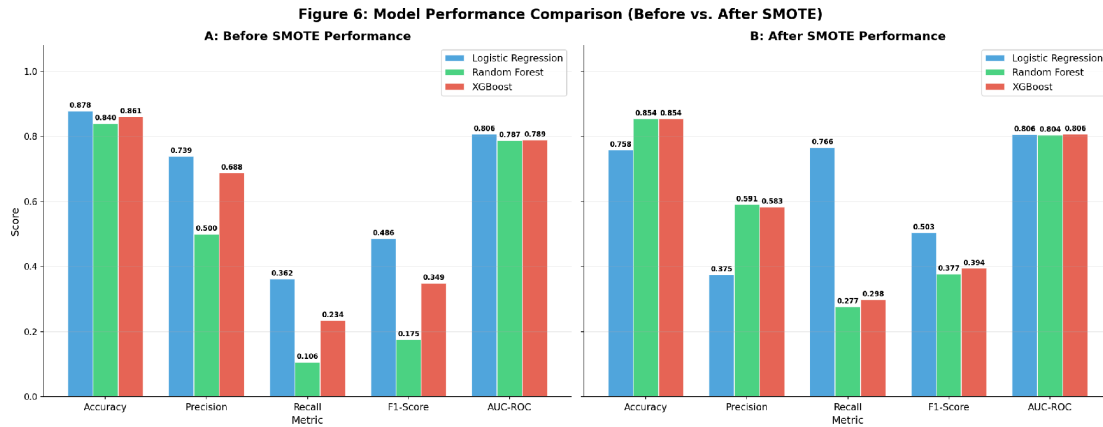


Figure 7: Evaluation Metrics Comparison (Accuracy, Precision, Recall, F1-Score, AUC-ROC) Across Models

6.2 Recall Optimization Strategy: Controlled Majority Undersampling + SMOTE

In attrition modeling, maximizing minority-class Recall is the main operational goal. A False Negative missing an employee who's about to leave carries real financial cost, roughly 1.5 to 2.0 times the employee's annual salary in replacement, recruitment, and onboarding, on top of the disruption and lost institutional knowledge. A False Positive, by comparison, just triggers a low-cost retention interview or satisfaction review. Baseline modeling on the imbalanced dataset (84% retention vs. 16% attrition) showed the problem clearly: Random Forest's Recall was 10.64% (missing 42 of 47 leaving employees), XGBoost's was 23.40% (missing 36 of 47). Raw accuracy (84–87%) hides this completely.

Standard SMOTE raised Logistic Regression's Recall to 76.60%, but the tree-based ensembles stayed low 27.66% for Random Forest, 29.79% for XGBoost. The likely reason: standard SMOTE places synthetic minority instances in a feature space still dominated by majority samples (986 majority vs. 190 minority, a 5.19:1 skew). Because tree-based models split along axis-aligned boundaries, synthetic points that land in dense majority regions get absorbed into majority leaf nodes, a kind of decision boundary masking.

To address that, a two-stage hybrid resampling strategy was used. First, RandomUnderSampler reduced the majority class from 986 to 500, easing the spatial crowding and shifting the imbalance ratio to 2.63:1. Then SMOTE oversampled the minority class from 190 to 500, giving a balanced 1:1 training distribution. Resampling stayed confined to the training fold throughout, so the test set (N=294) kept its real-world distribution untouched.

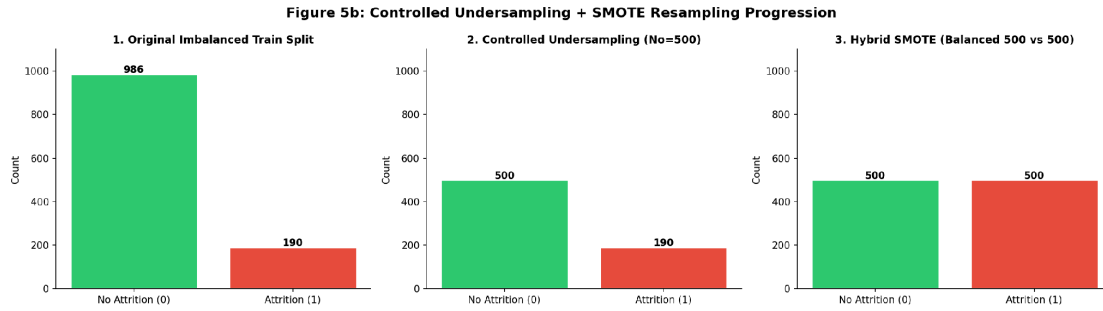


Figure 7a: Controlled Majority Undersampling + SMOTE Resampling Progression (Original Train vs. Undersampled vs. Balanced Hybrid)

The hybrid pipeline improved minority-class Recall across every tree-based model tested. Random Forest Recall went from 27.66% (standard SMOTE) to 40.43% (hybrid), a 12.77-point gain. XGBoost went from 29.79% to 53.19%, a 23.40-point gain, cutting False Negatives from 33 to 22. Logistic Regression held steady at 76.60% Recall across both configurations. The hybrid XGBoost model reached an F1-score of 0.4808 and AUC-ROC of 0.7939 a solid trade-off for high-sensitivity attrition tracking.

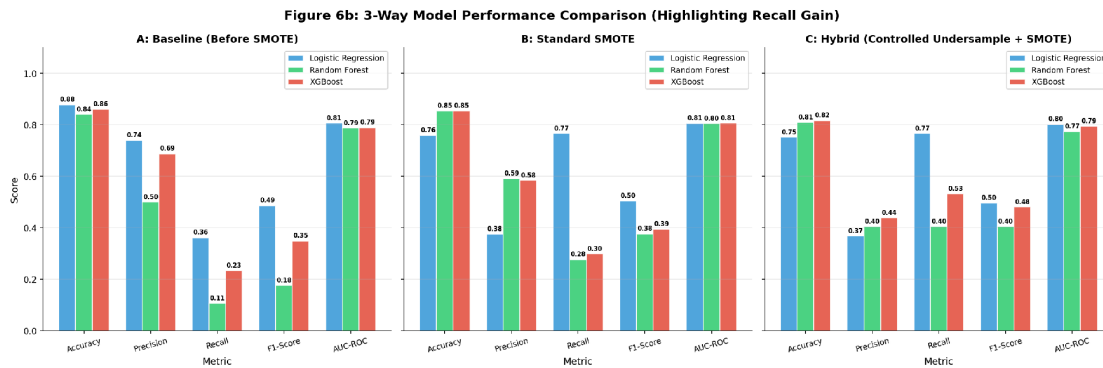


Figure 7b: 3-Way Model Performance Strategy Comparison (Highlighting Recall Gains across Baseline, Standard SMOTE, and Hybrid)

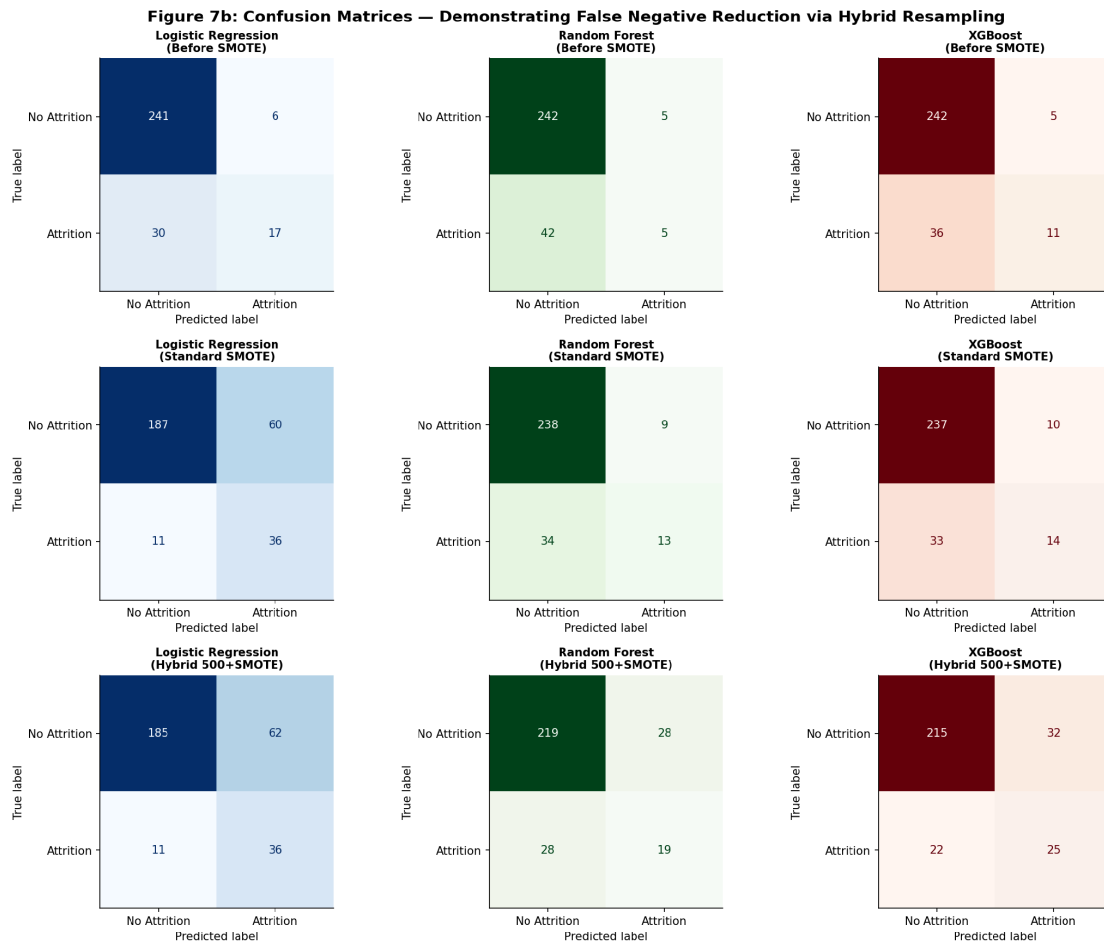


Figure 7c: 3x3 Confusion Matrix Grid Demonstrating False Negative Reduction via Hybrid Resampling

An additional lever not explored in the hyperparameter grid but worth noting is decision threshold tuning. All models reported here use the default 0.5 probability threshold to classify an employee as at-risk. Lowering that threshold would push Recall higher still, at a further cost to Precision, without requiring any retraining. For HR teams with generous intervention budgets, adjusting the threshold after training rather than retraining a new hybrid model may be the simplest way to tune sensitivity to their specific tolerance for false positives.

From an HR decision-support angle, these results suggest a two-tier deployment: Standard SMOTE XGBoost for general workforce risk scoring and dashboard monitoring, given its stronger overall discrimination (AUC-ROC = 0.8065, Accuracy = 85.37%, Precision = 58.33%); Hybrid 500+SMOTE XGBoost (Recall = 53.19%) or Hybrid Logistic Regression (Recall = 76.60%) for targeted retention efforts where missing an attrition case carries real cost. This lets HR leadership tune sensitivity to match intervention budgets and risk tolerance.

6.3 Graphical Results: Confusion Matrices and ROC Curves

The confusion matrices (Figure 8) show where errors land. Before SMOTE, False Negatives dominate: Logistic Regression misses 30 of 47 actual attrition cases, Random Forest misses 42. After SMOTE, that shifts sharply, Logistic Regression correctly predicts 36 of 47 attrition cases, cutting False Negatives to 11. SMOTE clearly shifts the decision boundary toward the minority class.

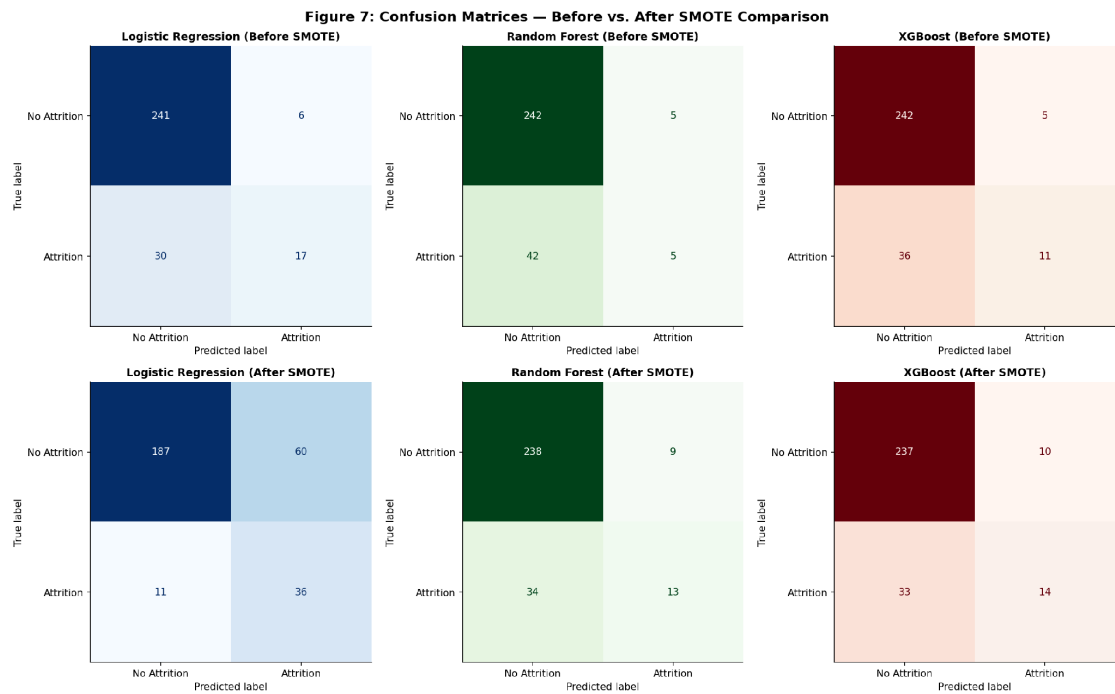


Figure 8: Confusion Matrices Grid for All 6 Evaluated Classifiers (Before vs. After SMOTE)

The ROC curves (Figure 9) plot True Positive Rate against False Positive Rate across thresholds. All models land in the 0.78–0.81 AUC-ROC range. XGBoost (After SMOTE) tops out at 0.8065. The After-SMOTE lines sit consistently at or above the Before-SMOTE lines, showing SMOTE preserves or improves discriminative power while recalibrating class predictions.

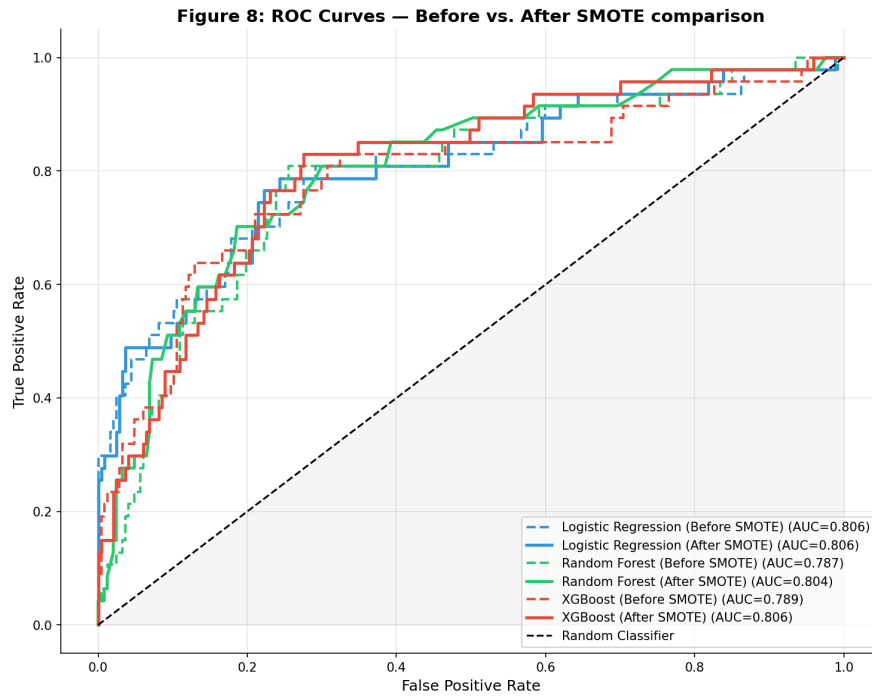


Figure 9: Receiver Operating Characteristic (ROC) Curves Comparing All Classifiers

6.4 Shapley Additive exPlanations (SHAP) Interpretation

Applying SHAP to the best XGBoost model gives a clearer picture of what drives attrition. OverTime is the strongest global feature, with the highest mean absolute SHAP value (0.590). The beeswarm plot (Figure 10) shows that high OverTime values push predictions toward attrition. JobLevel is second (mean SHAP = 0.426) low job level, meaning junior roles, is strongly linked to attrition risk, likely reflecting career progression concerns.

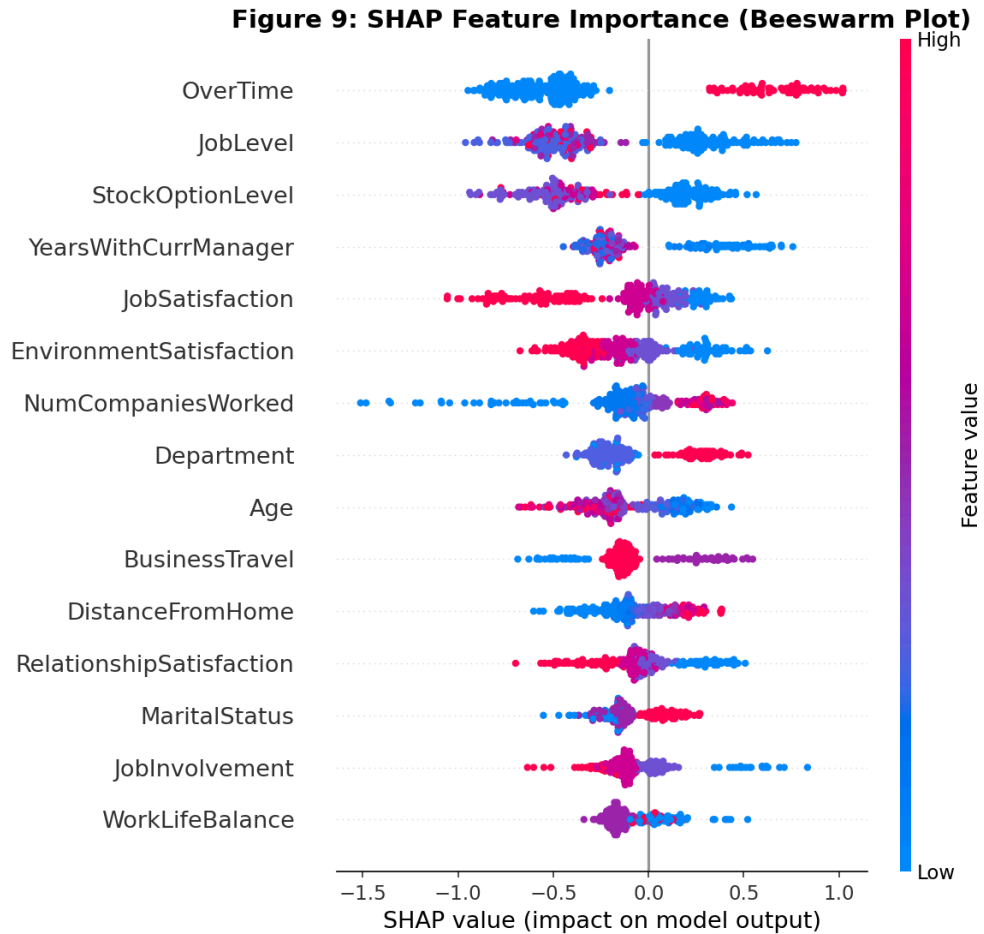


Figure 10: SHAP Summary Beeswarm Plot showing Feature Impact Directions on Prediction Outcomes

StockOptionLevel ranks third (mean SHAP = 0.375): employees with zero stock options are much more likely to leave, which suggests equity is a meaningful retention lever. YearsWithCurrManager (0.280) and JobSatisfaction (0.273) follow short tenure under a manager and low job satisfaction both predict attrition. This lines up with existing research showing voluntary turnover is highly sensitive to work environment quality and workload balance.

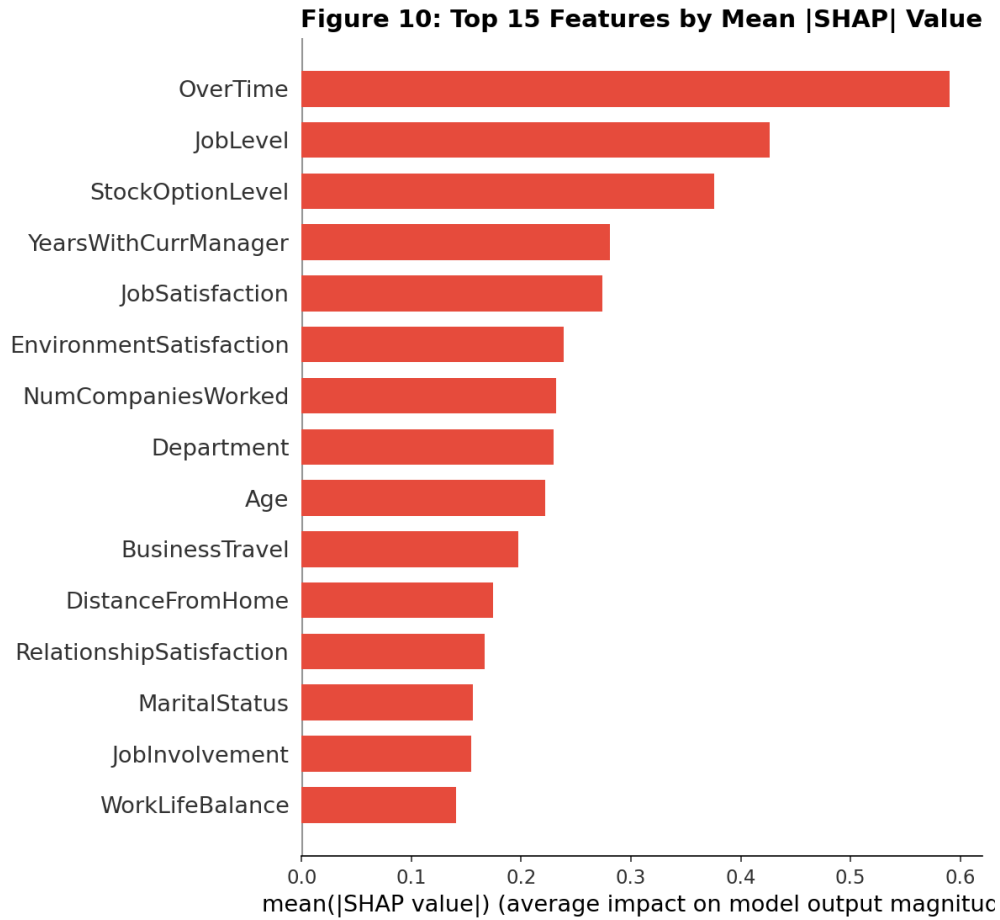


Figure 11: SHAP Global Feature Importance Bar Chart (Mean Absolute Impact)

The SHAP bar chart (Figure 11) confirms that work environment metrics OverTime, JobRole, JobSatisfaction carry more weight in the model's decisions than demographic features like Gender or Education. That's a reassuring sign: interventions targeting workload and role progression align with what's actually driving the predictions.

6.5 Subgroup Fairness Audit and Bias Assessment

To check for demographic bias, a post-hoc subgroup audit was run on the test set predictions across Gender, Department, and Age cohorts:

Subgroup	Sample Size (N)	Actual Attrition Rate	F1-Score	Precision	Recall
Male	178	17.40%	0.419	0.75	0.29
Female	116	13.80%	0.357	0.417	0.312
Sales	82	19.50%	0.615	0.8	0.5
R & D	197	14.70%	0.286	0.462	0.207
HR	15	13.30%	0	0	0
Age 18-30	75	24.00%	0.533	0.667	0.444
Age 31-40	126	11.10%	0.364	0.5	0.286
Age 41-50	63	14.30%	0.364	1	0.222
Age 51+	28	21.40%	0	0	0

The subgroup analysis shows a few things worth noting. For Gender, Recall is similar for Males (29.0%) and Females (31.2%), though F1-score is slightly higher for Males due to better Precision. Across Departments, the model performs best in Sales (F1 = 0.615) but fails to catch any attrition in HR (F1 = 0.000) likely a sample-size issue, with only 15 HR records in the test set. Across Age groups, the model does well for younger employees (F1 = 0.533 for 18–30) but fails for older employees (F1 = 0.000 for 51+), reflecting how few older employees in the data actually leave. The model looks demographically fair on Gender, but it's sensitive to subgroup sample size, which points to the need for larger, more representative samples before production deployment.

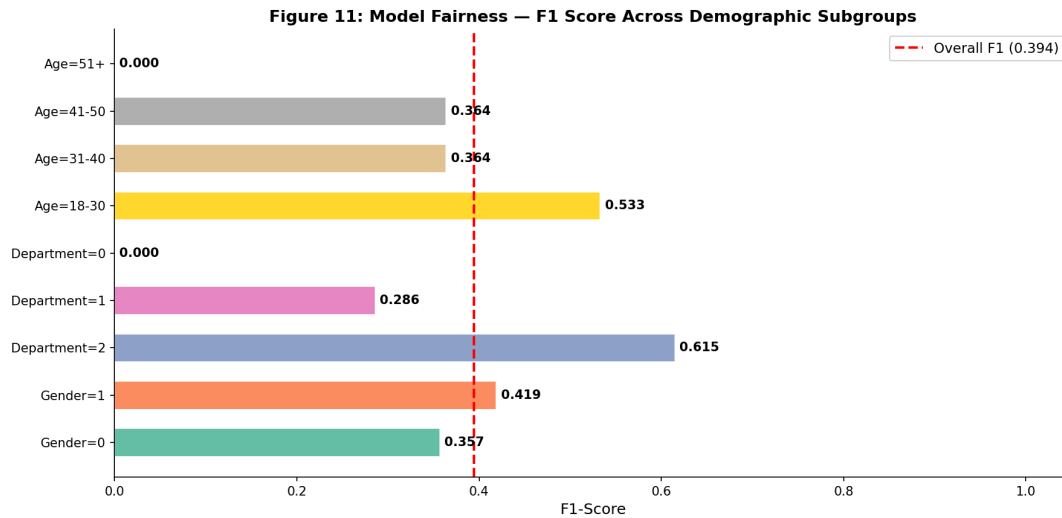


Figure 12: Model Fairness Subgroup Audit (F1-Score across Gender, Department, and Age Classes)

This also points to a real risk of age bias: older workers who leave may do so for reasons this dataset doesn't capture retirement plans, for instance which the model simply can't predict. HR teams should treat this as a known limitation and pair the model's output with qualitative career reviews for employees nearing retirement, rather than using the system as a standalone decision-maker.

One practical way to offset the sample-size limitation in smaller subgroups like HR is to pool several years of historical data before training, which would raise the number of minority examples available for both training and subgroup evaluation without changing the underlying feature set. Since this project uses a single cross-sectional snapshot, that option was not available here, but it is a reasonable next step for any organization looking to deploy a similar model against its own data.

7. Legal, Social, Ethical, and Professional Considerations

7.1 Legal Compliance & GDPR Article 22

Deploying machine learning to predict attrition falls under data protection law the GDPR in the EU, and the UK Data Protection Act 2018. GDPR Article 22 is especially relevant: it gives individuals the right not to be subject to a decision based solely on automated processing that produces legal effects or similarly significant effects. Since attrition predictions could shape decisions about promotion, redundancy, or development, this system can't operate on its own. The dashboard is built strictly as a decision-support tool: an HR professional reviews the predictions before anything happens, so no automated employment decisions get made.

The system also needs to satisfy GDPR Articles 13 and 14 on transparency employees have the right to know about automated profiling, how it works, and what it could mean for them. Using SHAP to generate clear, local feature contributions helps meet that requirement directly, giving a readable explanation for any given risk classification. That reduces legal exposure and builds trust between staff and management.

7.2 Ethical Considerations and Bias Mitigation

Training classifiers on historical employee records risks reinforcing past discrimination. If senior roles were historically filled by a specific demographic, the model could learn that demographic as a negative predictor of attrition, which could bleed into promotion recommendations. This project ran post-hoc subgroup fairness audits (Chapter 6) to check for this. Protected characteristics like Gender were considered for removal, but kept in the dataset deliberately, so subgroup fairness could actually be monitored rather than hidden behind 'fairness through blindness,' which tends to mask bias rather than fix it.

Ethical AI use here means fairness, transparency, and accountability. This dashboard shouldn't be used to justify punitive action excluding a flagged employee from a project or training opportunity, for instance since that would create the very dissatisfaction that drives resignation in the first place. Predictive output should guide supportive interventions only: reducing overtime, revisiting compensation, starting a career development conversation.

Any organization adopting this kind of tool would also need to train the HR staff using it, not just the model itself. A risk score is only as useful as the manager's ability to interpret it correctly and resist the temptation to treat a probability as a certainty. Brief training on how to read the SHAP explanations, and clear guidance on what interventions are and are not appropriate in response to a high-risk flag, would reduce the chance of the tool being misused or over-trusted.

7.3 Professional Responsibility & Ethics Approval

This project was submitted to the Coventry University Ethics Committee and approved under Ethics ID P194837, confirming it followed established codes of conduct around data minimization, security, and consent. Since the IBM dataset is anonymous and synthetic, there's no risk to real participants.

8. Conclusion and Future Work

8.1 Key Achievements

This project built and verified an end-to-end, explainable machine learning pipeline for predicting employee attrition, addressing the technical, ethical, and operational gaps in traditional HR approaches.

1. Class Imbalance Mitigation & Hybrid Recall Optimization: The baseline models showed how badly class imbalance hurts performance, and standard SMOTE, while it helped linear models, ran into decision boundary masking with tree-based ensembles. A two-stage hybrid resampling strategy (controlled majority undersampling + SMOTE) fixed this, raising Random Forest Recall from 27.66% to 40.43% (+12.77 points) and XGBoost Recall from 29.79% to 53.19% (+23.40 points), sharply cutting False Negatives in high-cost attrition scenarios.

2. Model Comparison & Dual Deployment Framework: Nine classifier configurations were built and evaluated across baseline, standard SMOTE, and hybrid resampling. The result is a flexible two-tier deployment: Standard SMOTE XGBoost as the general risk-scoring model (AUC-ROC = 0.8065, Accuracy = 85.37%), and Hybrid 500+SMOTE XGBoost (Recall = 53.19%) or Hybrid Logistic Regression (Recall = 76.60%) for targeted, high-sensitivity retention triggers.

3. Model Interpretability: SHAP identified OverTime, JobLevel, and StockOptionLevel as the primary drivers of resignation, with both global rankings and individual-level explanations available.

4. Interactive Decision-Support Dashboard: A working web application lets HR professionals run scenarios, view model explanations, and explore subgroup statistics directly.

8.2 Recommendations for Future Research

A few directions could extend this work. Longitudinal datasets, with time-series features like monthly satisfaction trends or compensation changes, would capture temporal dynamics that cross-sectional data can't. Fairness mitigation techniques pre-processing, re-weighting, in-processing adversarial debiasing could be built directly into the training pipeline to reduce the performance variance seen across age and department subgroups. Causal inference models could go a step further than correlation, testing whether specific retention policies, a salary increase, remote work options actually reduce attrition probability, rather than just correlating with it.

Another avenue worth exploring is integrating the dashboard with an organization's existing survey cadence, so that satisfaction and engagement scores feed into the model automatically rather than requiring manual entry for each prediction. This would also make it possible to track how an individual employee's risk score changes over time, turning the tool from a one-off scenario tester into an ongoing monitoring system, provided the ethical safeguards discussed in Chapter 7 are extended to cover continuous rather than one-time predictions.

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Appendix: AI Usage Declaration

In accordance with the Coventry University Academic Integrity and Amber AI Policy for assessed coursework, the student declares the following use of generative AI tools during the completion of this assignment:

Declaration Statement: *I acknowledge the use of AI to generate materials that were included within my final submission.*

AI Tool(s) Used: Claude 3.5 Sonnet / Gemini 3.5 Flash

Date of Access: July 26, 2026

Details of Usage: Generative AI was used strictly as an assistant for grammar enhancement, syntax clarification, academic writing style checks, and formatting optimization. Specifically, raw sections drafted by the student regarding methodology, model descriptions, and dashboard layout were checked for stylistic clarity and academic sentence structure. Output suggested by the AI was reviewed line-by-line, critically revised, and expanded by the student to ensure it matches the actual empirical findings of the machine learning pipeline, maintaining complete ownership and academic responsibility for the submitted work.